PHYSICS

Higher Level

Set A | Paper 1A and 1B

Challenge mock examination

Time allowed: 120 minutes Maximum mark: 60

Candidate name: _____

Candidate number: _____

Date: _____

Instructions

• A calculator and a clean current Physics data booklet are permitted throughout.

• Paper 1A and Paper 1B share one uninterrupted examination time. Paper 1A has 40 multiple-choice questions; Paper 1B has two compulsory experimental/data questions worth 20 marks altogether.

• Answer all questions. Show physical reasoning, working and units in written responses.

• Use the constants and assumptions given in a question. Otherwise use the data booklet and retain unrounded intermediate values.

• Give numerical answers to an appropriate number of significant figures. Diagrams are schematic unless explicitly stated to be to scale.

• The number in square brackets gives the maximum mark for a written part.

Thriving Scholars challenge practice | Current Physics syllabus | 2026

Paper 1A | Question paper

Paper 1A - Multiple choice

Select one answer, A, B, C or D, for each question. Use the answer grid or record your choices on the portal.

1	2	3	4	5	6	7	8	9	10

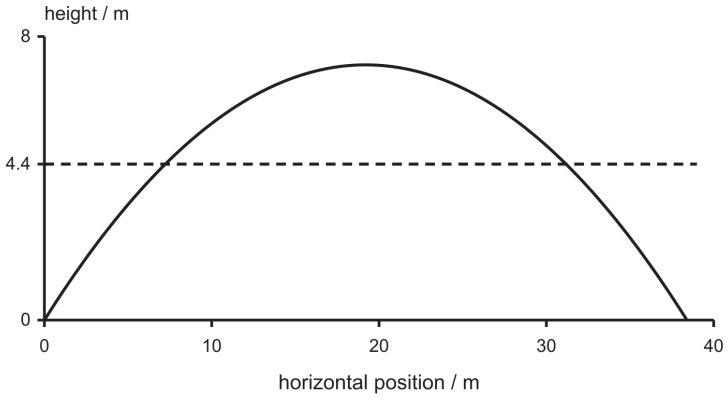
11	12	13	14	15	16	17	18	19	20

21	22	23	24	25	26	27	28	29	30

31	32	33	34	35	36	37	38	39	40

1A Question 1 [1 mark]

A test capsule is launched with horizontal and vertical velocity components 16 and $12\,\mathrm{m\,s^{-1}}$. It crosses the horizontal plane $4.4\,\mathrm{m}$ above launch twice. With $g = 10\,\mathrm{m\,s^{-2}}$ and no drag, what is the horizontal separation of these crossings?



A. $12.0\,\mathrm{m}$

B. $24.0\,\mathrm{m}$

C. $38.4\,\mathrm{m}$

D. $19.2\,\mathrm{m}$

1A Question 2 [1 mark]

A cart has speed $3.0\,\mathrm{m\,s^{-1}}$ at $x = 0$ and $5.0\,\mathrm{m\,s^{-1}}$ at $x = 4.0\,\mathrm{m}$. Its acceleration is constant. What is its speed at $x = 16.0\,\mathrm{m}$?

A. $10.0\,\mathrm{m\,s^{-1}}$

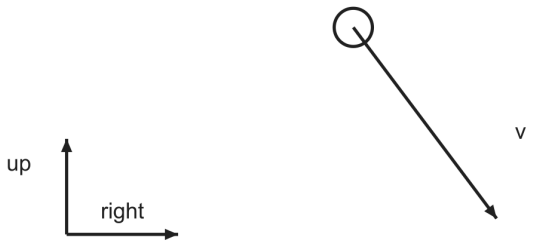
B. $7.00\,\mathrm{m\,s^{-1}}$

C. $8.54\,\mathrm{m\,s^{-1}}$

D. $11.0\,\mathrm{m\,s^{-1}}$

1A Question 3 [1 mark]

In an idealized motion model, a test body has linear drag $\mathbf{F}_d = -b\mathbf{v}$ and terminal downward speed $20\,\mathrm{m\,s^{-1}}$. At one instant its rightward velocity is $12\,\mathrm{m\,s^{-1}}$ and downward velocity is $16\,\mathrm{m\,s^{-1}}$. Ignore buoyancy and use $g = 10\,\mathrm{m\,s^{-2}}$. What is the acceleration magnitude?



A. $2.00\,\mathrm{m\,s^{-2}}$

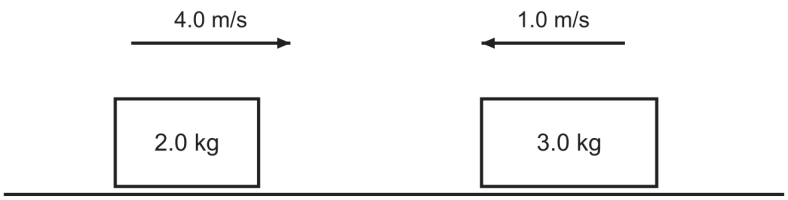
B. $6.32\,\mathrm{m\,s^{-2}}$

C. $14.1\,\mathrm{m\,s^{-2}}$

D. $20.0\,\mathrm{m\,s^{-2}}$

1A Question 4 [1 mark]

A $2.0\,\text{kg}$ cart moves right at $4.0\,\text{m}\,\text{s}^{-1}$ towards a $3.0\,\text{kg}$ cart moving left at $1.0\,\text{m}\,\text{s}^{-1}$. They latch together on a level frictionless track. What fraction of their initial kinetic energy is dissipated?



A. $6/7$

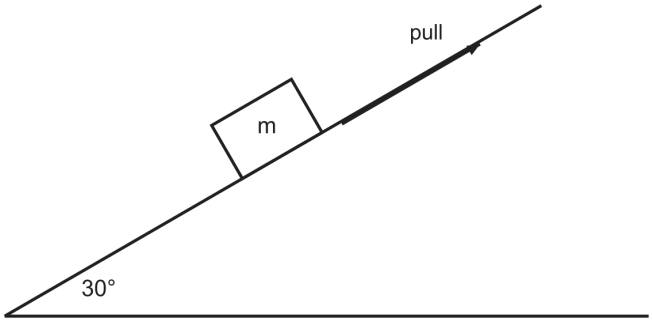
B. $1/7$

C. $3/7$

D. $5/7$

1A Question 5 [1 mark]

A $2.0\,\text{kg}$ package is pulled up a 30° ramp at constant speed by a force parallel to the ramp. The kinetic-friction coefficient is $1/(2\sqrt{3})$. With $g = 10\,\text{m}\,\text{s}^{-2}$, what pulling force is required?



A. $15\,\text{N}$

B. $20\,\text{N}$

C. $5.0\,\text{N}$

D. $10\,\text{N}$

1A Question 6 [1 mark]

A lift raises an $80\,\text{kg}$ load through $20\,\text{m}$, consuming $25\,\text{kJ}$ of electrical energy. During steady lowering through the same height, it returns 50% of the lost gravitational energy to the supply. Ignore other moving masses and use $g = 10\,\text{m}\,\text{s}^{-2}$. What net electrical energy is imported per cycle?

A. $33\,\text{kJ}$

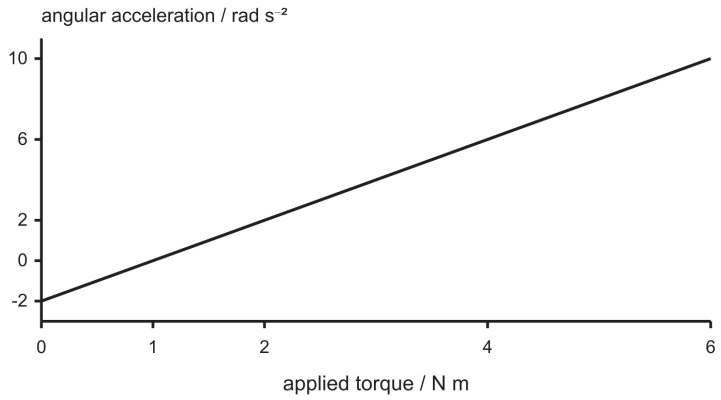
B. $9.0\,\text{kJ}$

C. $17\,\text{kJ}$

D. $8.0\,\text{kJ}$

1A Question 7 [1 mark]

The graph relates angular acceleration to the applied driving torque for a flywheel. Bearing friction supplies a constant opposing torque. What are the moment of inertia I and frictional torque τ_f?



A. $I = 0.50\,\text{kg}\,\text{m}^2,\ \tau_f = 2.0\,\text{N}\,\text{m}$

B. $I = 2.0\,\text{kg}\,\text{m}^2,\ \tau_f = 2.0\,\text{N}\,\text{m}$

C. $I = 0.50\,\text{kg}\,\text{m}^2,\ \tau_f = 1.0\,\text{N}\,\text{m}$

D. $I = 2.0\,\text{kg}\,\text{m}^2,\ \tau_f = 1.0\,\text{N}\,\text{m}$

1A Question 8 [1 mark]

A beam of muons travels at $0.80c$ through a $480\,\text{m}$ detector measured in the laboratory. Their mean proper lifetime is $2.0\,\mu\text{s}$. The surviving fraction is $\exp(-\Delta\tau/\tau_0)$, where $\Delta\tau$ is elapsed proper time. What fraction survives?

A. 0.549

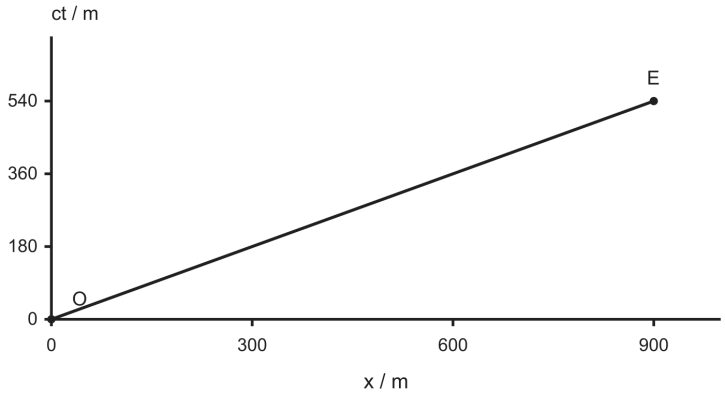
B. 0.189

C. 0.698

D. 0.368

1A Question 9 [1 mark]

Events O and E have separations $\Delta x = 900\,\text{m}$ and $c\Delta t = 540\,\text{m}$ in frame S. Frame $S\prime$ moves at $+0.60c$ along x. What spatial separation does $S\prime$ assign to these two events?



A. $720\,\text{m}$

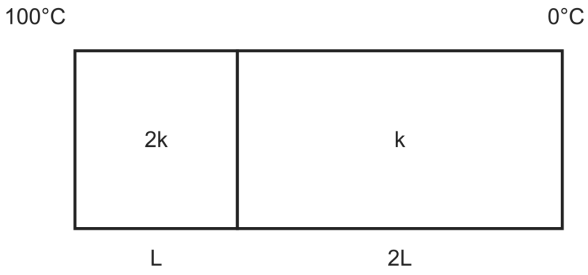
B. $432\,\text{m}$

C. $1125\,\text{m}$

D. $576\,\text{m}$

1A Question 10 [1 mark]

Two large slabs of equal area are in perfect thermal contact. The left slab has thickness L and conductivity $2k$; the right has thickness $2L$ and conductivity k. Their outer surfaces are maintained at $100\,^\circ\mathrm{C}$ and $0\,^\circ\mathrm{C}$. In steady one-dimensional conduction, what is the interface temperature?



Cross section; equal face areas

A. $80\,^\circ\mathrm{C}$

B. $67\,^\circ\mathrm{C}$

C. $20\,^\circ\mathrm{C}$

D. $50\,^\circ\mathrm{C}$

1A Question 11 [1 mark]

In a spherical airless-planet model, equilibrium satisfies $T^4 \propto (1-\alpha)/\varepsilon$ at fixed incident stellar flux. A surface treatment changes albedo from 0.20 to 0.60 and infrared emissivity from 1.00 to 0.50. What is $T_{\mathrm{new}}/T_{\mathrm{old}}$?

A. $2^{-1/4}$

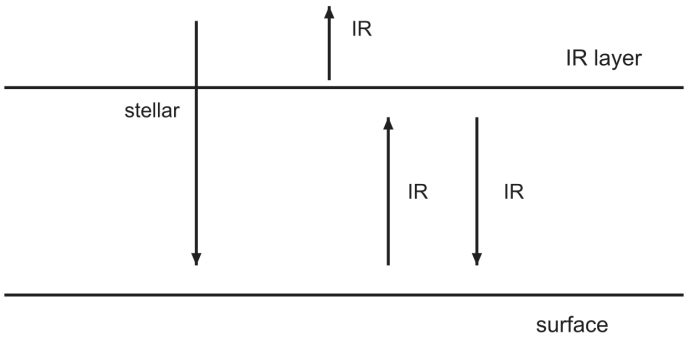
B. 1

C. $\sqrt{2}$

D. $1/\sqrt{2}$

1A Question 12 [1 mark]

A planet absorbs a global-average stellar flux F. Add one atmospheric layer transparent to stellar radiation, perfectly absorbing at infrared wavelengths and emitting infrared equally upward and downward. Surface emissivity is one. What is the ratio of the new surface temperature to its airless value?



A. 2

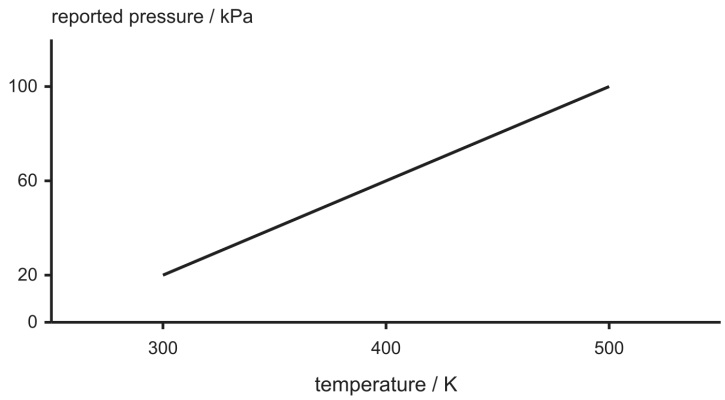
B. 1

C. $\sqrt{2}$

D. $2^{1/4}$

1A Question 13 [1 mark]

A fixed amount $0.200\,\text{mol}$ of ideal gas is held at constant volume. A sensor reports gauge pressure with an unknown constant zero offset. Its graph has gradient $400\,\text{Pa}\,\text{K}^{-1}$. With $R = 8.31\,\text{J}\,\text{mol}^{-1}\,\text{K}^{-1}$, what is the gas volume?



A. $8.31\,\text{L}$

B. $1660\,\text{L}$

C. $2.08\,\text{L}$

D. $4.16\,\text{L}$

1A Question 14 [1 mark]

Equal volumes of monatomic ideal neon and argon are at the same pressure and temperature. An argon atom has twice the mass of a neon atom. Which row gives $U_{\mathrm{Ar}}/U_{\mathrm{Ne}}$ and $v_{\mathrm{rms,Ar}}/v_{\mathrm{rms,Ne}}$?

A. $1,\ \sqrt{2}$

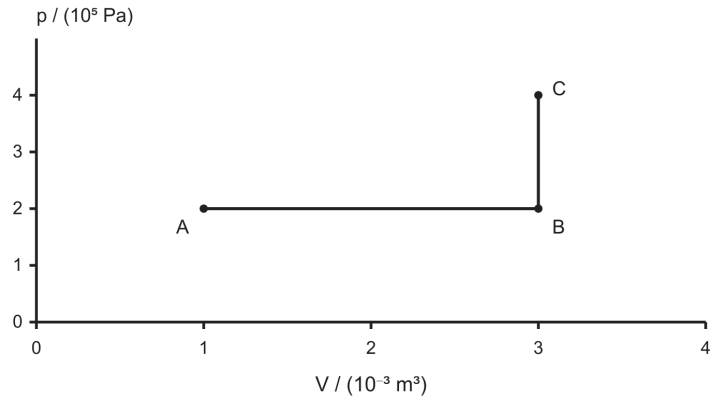
B. $2,\ 1/\sqrt{2}$

C. $1,\ 1/\sqrt{2}$

D. $1/2,\ 2$

1A Question 15 [1 mark]

A monatomic ideal gas follows $A \to B \to C$ on the graph. Horizontal coordinates are volume in $10^{-3}\,\mathrm{m}^3$ and vertical coordinates pressure in $10^5\,\mathrm{Pa}$. What total thermal energy is supplied to the gas?



A. $1100\,\mathrm{J}$

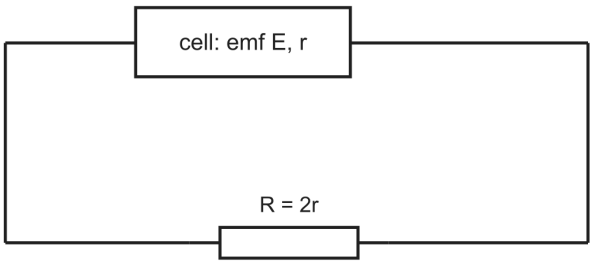
B. $1500\,\mathrm{J}$

C. $2300\,\mathrm{J}$

D. $1900\,\mathrm{J}$

1A Question 16 [1 mark]

A cell has emf $\mathcal{E}$ and fixed internal resistance r. A resistor $R = 2r$ is connected. What fraction of the maximum possible external-resistor power is dissipated in this resistor, when R may otherwise be any positive value?



A. $3/4$

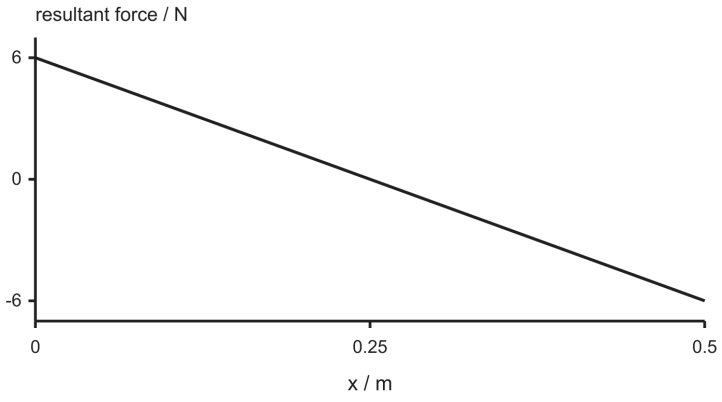
B. $8/9$

C. $2/3$

D. 1

1A Question 17 [1 mark]

A $0.600\,\text{kg}$ slider experiences the resultant force $F = 6.0 - 24x$ in SI units over the shown range. What is the period of small oscillations?



A. $0.497\,\text{s}$

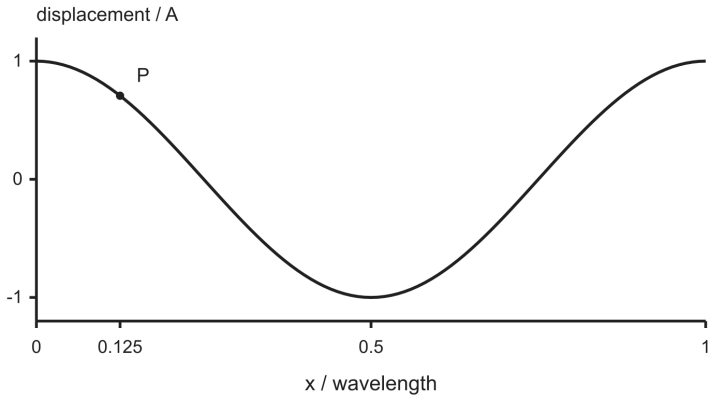
B. $1.99\,\text{s}$

C. $0.993\,\text{s}$

D. $0.158\,\text{s}$

1A Question 18 [1 mark]

The snapshot is a transverse wave travelling towards increasing x. Point P is at $x = \lambda/8$ and the snapshot is $y = A\cos(2\pi x/\lambda)$. At this instant, what are the signs of the particle velocity and acceleration at P, taking upward as positive?



A. velocity negative; acceleration positive

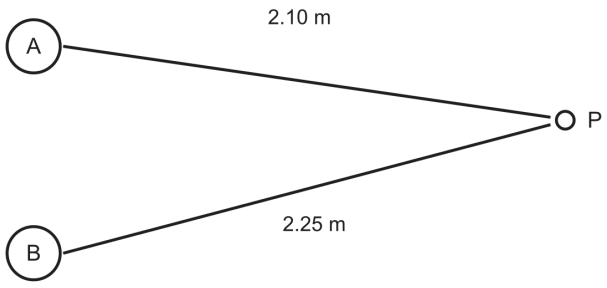
B. velocity positive; acceleration positive

C. velocity negative; acceleration negative

D. velocity positive; acceleration negative

1A Question 19 [1 mark]

Two coherent sound sources emit at wavelength $0.60\,\text{m}$. Source B leads source A by phase $\pi/2$. Their path lengths to P are $r_A = 2.10\,\text{m}$ and $r_B = 2.25\,\text{m}$. Propagation contributes phase $-2\pi r/\lambda$. What is the arrival phase difference at P, modulo 2π?



A. $\pi/2$

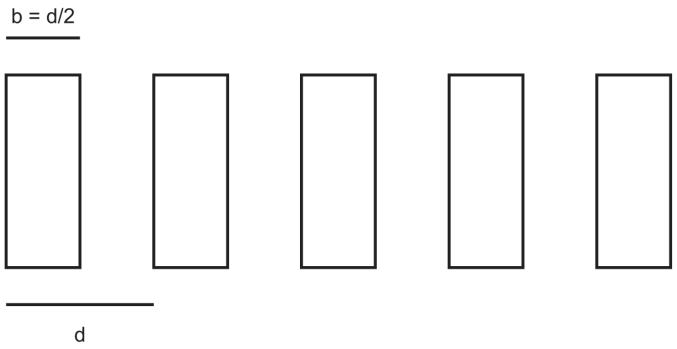
B. $3\pi/2$

C. π

D. 0

1A Question 20 [1 mark]

Monochromatic light of wavelength λ is normally incident on a grating with separation $d = 4.4\lambda$ and rectangular slit width $b = d/2$. Principal orders satisfy $d\sin\theta = n\lambda$; envelope minima satisfy $b\sin\theta = m\lambda$ for nonzero integer m. How many principal maxima remain after coincident envelope minima are excluded?



A. 8

B. 9

C. 4

D. 5

1A Question 21 [1 mark]

An open-open pipe has adjacent resonance frequencies $180\ \text{Hz}$ and $270\ \text{Hz}$. One end is then closed without changing its length or the sound speed. Ignoring end corrections, what is the lowest resonance frequency of the new pipe?



A. $60\ \text{Hz}$

B. $90\ \text{Hz}$

C. $45\ \text{Hz}$

D. $180\ \text{Hz}$

1A Question 22 [1 mark]

A lightly damped mechanical oscillator is driven by a sinusoidal force of fixed amplitude. Its mass and spring are unchanged, but the damping is increased while a resonance peak remains. Which change is expected in the steady-state amplitude-frequency graph?

A. unchanged peak height, shifted to lower frequency

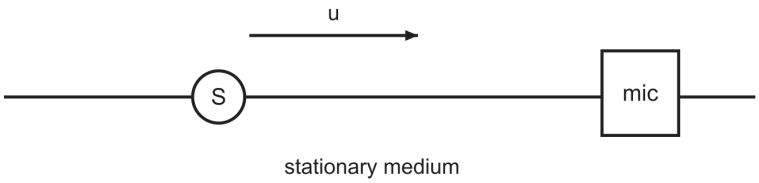
B. taller and narrower peak, shifted to higher frequency

C. lower and narrower peak, shifted to higher frequency

D. lower and broader peak, shifted to lower frequency

1A Question 23 [1 mark]

A stationary microphone records a source approaching directly and then receding directly at the same constant speed. The approaching frequency is twice the receding frequency. The medium is stationary. What is the source speed as a fraction of the sound speed v?



A. $v/4$

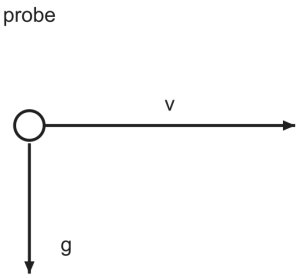
B. $v/3$

C. $2v/3$

D. $v/2$

1A Question 24 [1 mark]

At one instant a probe velocity is horizontal while the gravitational field at its position is vertical downward. No other force acts. Which statement is correct at that instant?



A. Kinetic energy must be decreasing because gravity acts.

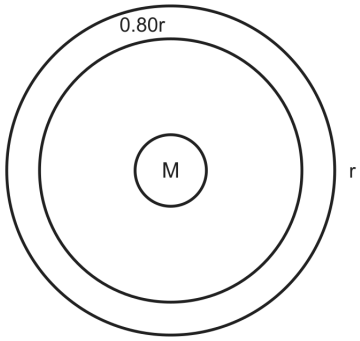
B. Kinetic energy has zero instantaneous rate of change, but momentum does not.

C. Momentum has zero instantaneous rate of change, but kinetic energy does not.

D. Kinetic energy and momentum are both constant in time.

1A Question 25 [1 mark]

A satellite slowly descends through nearly circular orbits from radius r to $0.80r$. Its mass stays constant. Let its initial kinetic energy be K. For a circular orbit the total energy is $-GMm/(2r)$. How much energy is dissipated during the descent?



A. $0.80K$

B. $0.25K$

C. $1.25K$

D. $0.20K$

1A Question 26 [1 mark]

Charges $+Q$ and $-4Q$ are fixed at $x = 0$ and $x = 3d$. At which coordinate is the resultant electric field zero?



A. $x = 6d$

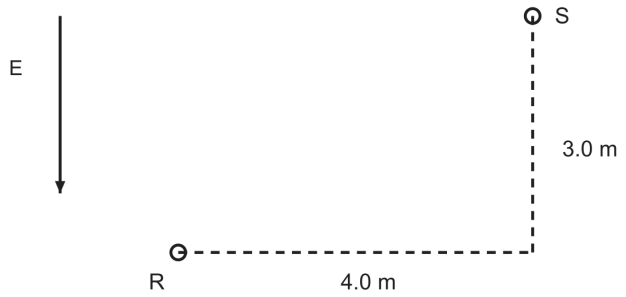
B. $x = d$

C. $x = -3d$

D. $x = -d$

1A Question 27 [1 mark]

A uniform electric field of strength $200\,\mathrm{V\,m^{-1}}$ is vertically downward. Point S is $3.0\,\mathrm{m}$ above and $4.0\,\mathrm{m}$ right of R. What external work is required to move a $+2.0\,\mathrm{nC}$ charge slowly from R to S?



A. $+1.2\,\mu\mathrm{J}$

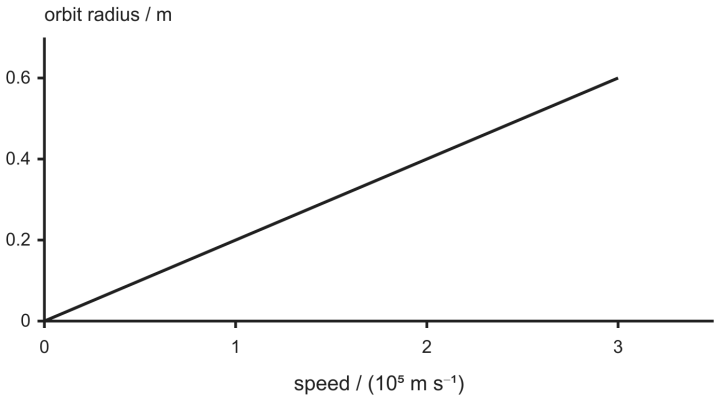
B. $-1.2\,\mu\mathrm{J}$

C. $-2.0\,\mu\mathrm{J}$

D. $+2.0\,\mu\mathrm{J}$

1A Question 28 [1 mark]

The graph shows the orbit radius of a nonrelativistic ion in a uniform field $B = 0.40\,\mathrm{T}$, perpendicular to its velocity. What is the ion's charge-to-mass ratio magnitude?



A. $1.25 \times 10^{-6}\,\mathrm{C\,kg^{-1}}$

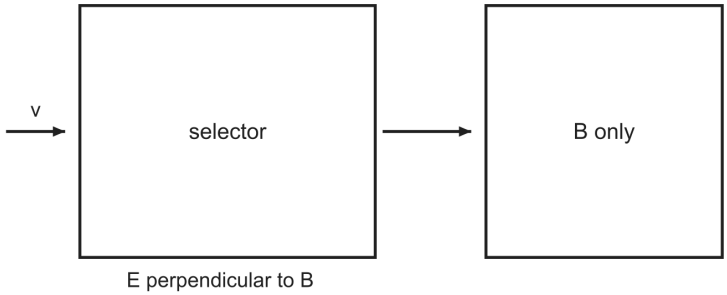
B. $1.25 \times 10^{6}\,\mathrm{C\,kg^{-1}}$

C. $1.25 \times 10^{5}\,\mathrm{C\,kg^{-1}}$

D. $5.00 \times 10^{6}\,\mathrm{C\,kg^{-1}}$

1A Question 29 [1 mark]

Protons and alpha particles both pass undeflected through the same perpendicular electric and magnetic fields. They then enter a second, purely magnetic region at right angles to its field. An alpha particle has four times a proton's mass and twice its charge. What is r_α / r_p in the second region?



A. 2

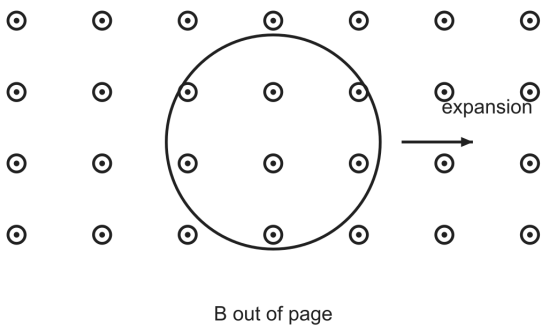
B. 4

C. 1

D. $1/2$

1A Question 30 [1 mark]

A flexible conducting ring lies in a steady uniform magnetic field directed out of the page. An external agent starts increasing its area. Which induced conventional current and magnetic-force tendency oppose the change?



A. clockwise current; tendency to expand

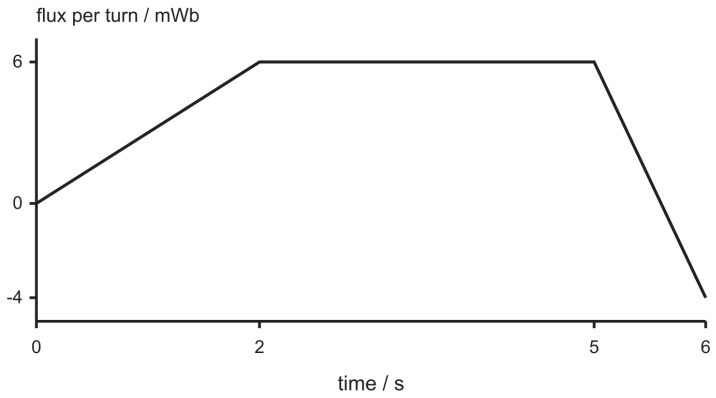
B. counterclockwise current; tendency to contract

C. clockwise current; tendency to contract

D. counterclockwise current; tendency to expand

1A Question 31 [1 mark]

A 50-turn coil has the flux per turn shown. Positive emf is defined by $\mathcal{E} = -N\Delta\Phi/\Delta t$. Over the full six seconds, what are the signed mean emf and the mean magnitude of emf?



A. $-0.0333\,\text{V},\ 0.133\,\text{V}$

B. $+0.0333\,\text{V},\ 0.133\,\text{V}$

C. $+0.133\,\text{V},\ 0.0333\,\text{V}$

D. $+0.200\,\text{V},\ 0.200\,\text{V}$

1A Question 32 [1 mark]

An alpha-scattering experiment is repeated with the same beam energy and metal, but a foil of half the thickness. Both foils are sufficiently thin that single nuclear scattering dominates. Which change is expected?

A. approximately half as many large-angle events; every particle loses half its kinetic energy

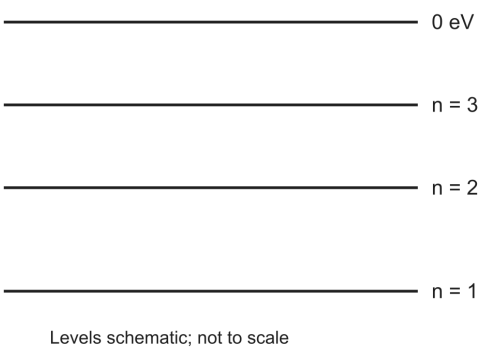
B. approximately twice as many large-angle events; most particles still pass nearly straight through

C. unchanged large-angle fraction; all transmitted particles have half their original speed

D. approximately half as many large-angle events; most particles still pass nearly straight through

1A Question 33 [1 mark]

Hydrogen has $E_n = -13.6/n^2\ \text{eV}$. What is the ratio of the energy required to ionize an atom initially in $n = 2$ to the energy of a photon emitted in $n = 3 \to 2$?



Levels schematic; not to scale

A. $9/5$

B. $5/9$

C. $36/5$

D. 4

1A Question 34 [1 mark]

A metal is illuminated above threshold by monochromatic light. Its wavelength is halved while total incident power and illuminated area remain fixed. Assume every incident photon releases one collected electron. Which change occurs?

A. saturation current halves; maximum kinetic energy is less than doubled

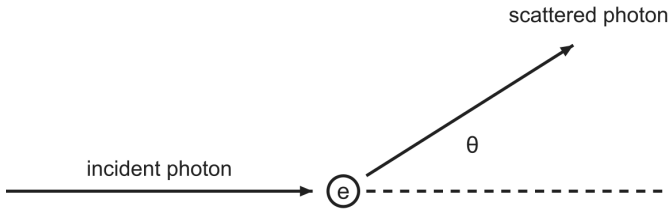
B. saturation current halves; maximum kinetic energy is more than doubled

C. saturation current is unchanged; maximum kinetic energy doubles

D. saturation current doubles; maximum kinetic energy doubles

1A Question 35 [1 mark]

A photon has initial wavelength equal to the electron Compton wavelength $\lambda_C = h/(m_e c)$. It scatters from an initially stationary free electron. What is the ratio of electron kinetic energy for photon scattering angle 180° to that for 90°?



A. 1

B. 2

C. $2/3$

D. $4/3$

1A Question 36 [1 mark]

Two nonrelativistic particles start from rest and accelerate through the same potential difference. Particle X has mass m and charge q; particle Y has mass $4m$ and charge $2q$, of the same sign. What is λ_Y/λ_X?

A. $1/4$

B. $\sqrt{2}$

C. $1/(2\sqrt{2})$

D. $1/2$

1A Question 37 [1 mark]

A detector reads 110, 70 and $50\,\mathrm{s}^{-1}$ at times 0, T and $2T$, where T is the source half-life. Background is constant and detector response is linear. What raw count rate is expected at $3T$?

A. $35\,\mathrm{s}^{-1}$

B. $45\,\mathrm{s}^{-1}$

C. $40\,\mathrm{s}^{-1}$

D. $30\,\mathrm{s}^{-1}$

1A Question 38 [1 mark]

One possible fission is $^{239}_{94}\mathrm{Pu} + {}^{1}_{0}n \to {}^{144}_{56}\mathrm{Ba} + {}^{93}_{38}\mathrm{Sr} + 3{}^{1}_{0}n$. The $^{93}\mathrm{Sr}$ fragment later becomes $^{93}_{40}\mathrm{Zr}$ through beta decays without other nuclear changes. What emissions are required?

A. one electron and one antineutrino

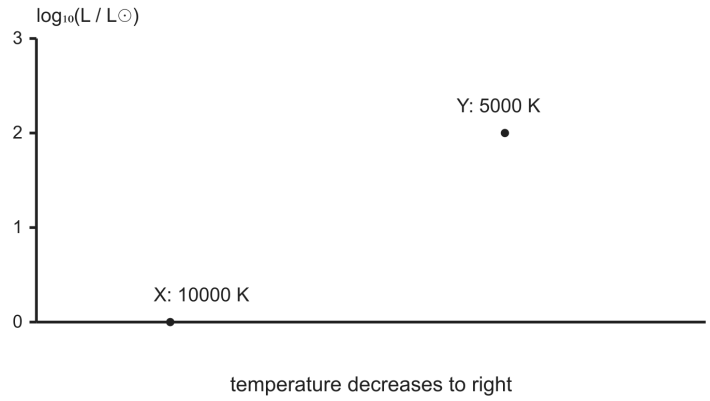
B. three electrons and three antineutrinos

C. two positrons and two neutrinos

D. two electrons and two antineutrinos

1A Question 39 [1 mark]

The HR diagram places star X at $(10000\,\mathrm{K}, 1L_\odot)$ and star Y at $(5000\,\mathrm{K}, 100L_\odot)$. Treat both as spherical black bodies. What is R_Y/R_X?



A. 40

B. 2.5

C. 0.40

D. 10

1A Question 40 [1 mark]

The maximum change in a star's apparent direction between observations six months apart is $0.080\,\mathrm{arcsec}$. Its apparent brightness is $6.0 \times 10^{-10}\,\mathrm{W\,m^{-2}}$. Using $1\,\mathrm{pc} = 3.086 \times 10^{16}\,\mathrm{m}$, what is the star's luminosity?

A. $4.49 \times 10^{27}\,\mathrm{W}$

B. $3.77 \times 10^{10}\,\mathrm{W}$

C. $1.79 \times 10^{28}\,\mathrm{W}$

D. $1.12 \times 10^{27}\,\mathrm{W}$

1B Question 1 | Question paper

Paper 1B - Experimental and data-based questions

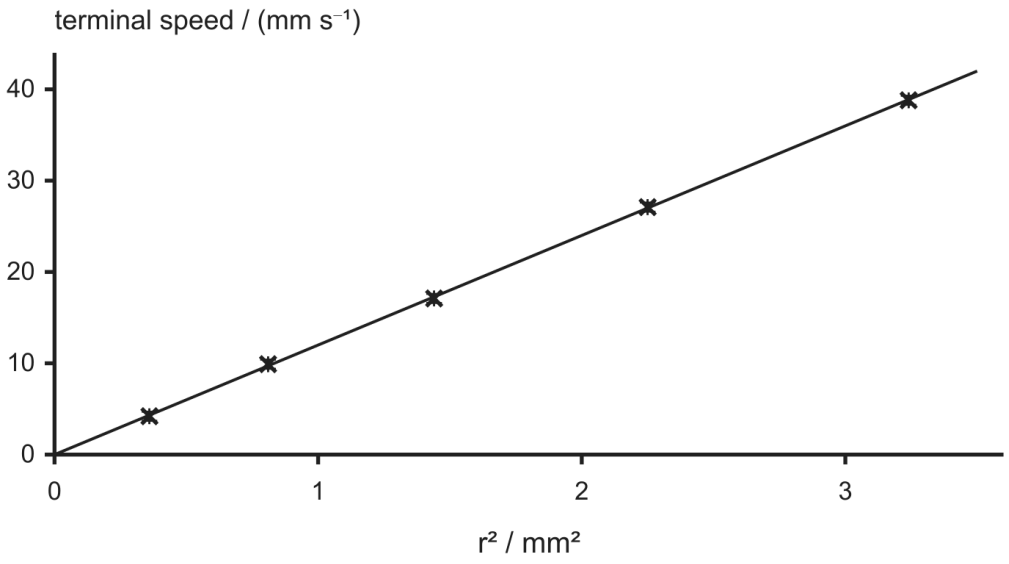
Continue within the same examination time. Answer both questions. Use the data, graphs and uncertainty information provided.

1B Question 1 [10 marks]

A model laboratory dataset is used to investigate a viscous liquid. A solid calibration body of known volume $4.00 \times 10^{-6}\,\mathrm{m}^3$ has balance readings $0.03120\,\mathrm{kg}$ in air and $0.02640\,\mathrm{kg}$ when fully immersed. Ignore air buoyancy. Use $g = 9.81\,\mathrm{m\,s^{-2}}$.

Spherical steel beads of density $\rho_s = 7800\,\mathrm{kg\,m^{-3}}$ settle through the same liquid. The plotted terminal speeds have uncertainty $\pm 0.5\,\mathrm{mm\,s^{-1}}$; radius-squared uncertainties are negligible. After assessing the graph, use the reported gradient $s = (12.0 \pm 0.4)\,(\mathrm{mm\,s^{-1}})/\mathrm{mm}^2$ for subsequent calculations. The independent density measurement gives $\rho_l = (1200 \pm 75)\,\mathrm{kg\,m^{-3}}$.

For terminal Stokes motion, $v_t = 2g(\rho_s - \rho_l)r^2/(9\eta)$. Treat g and ρ_s as exact. A second test examines the validity of the Stokes model using the supplied diagnostic $\mathrm{Re} = 2\rho_l r v_t/\eta$; the model is more reliable when $\mathrm{Re} \ll 1$.



(a) Calculate the liquid density from the two balance readings and the calibration-body volume. [2]

(b) Explain why the graph is approximately consistent with $v_t \propto r^2$, and convert the reported gradient to SI units. [2]

(c) Determine η and its absolute uncertainty by adding the relevant fractional uncertainties. Give suitable units and precision. [3]

(d) Use the largest bead to evaluate the Stokes assumption. Suggest one change that reduces this limitation, and explain why terminal speed must be checked separately. [3]

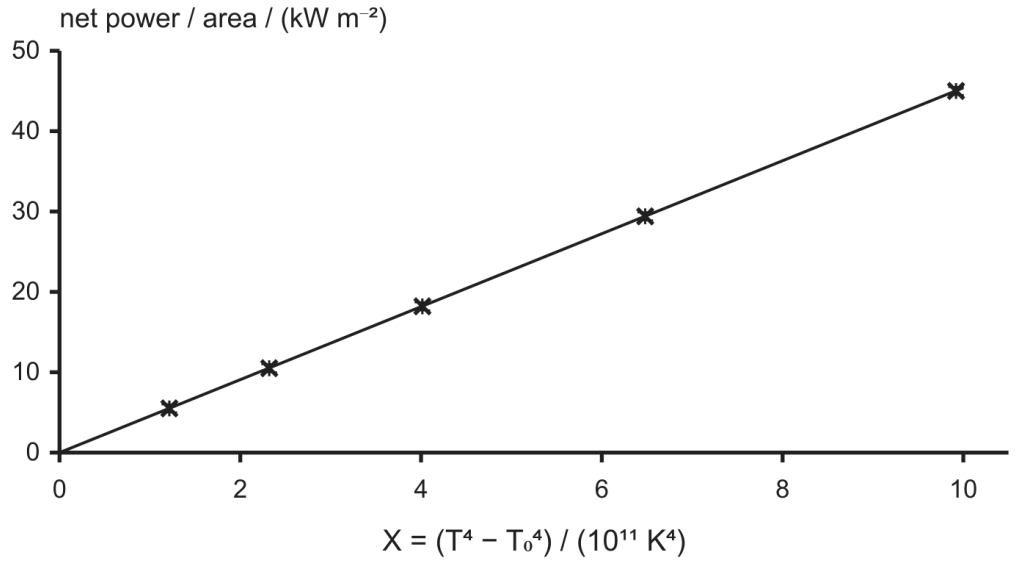
1B Question 2 | Question paper

1B Question 2 [10 marks]

A model vacuum-chamber investigation tests radiative power from a coated tile. Its area is known and its emissivity is assumed temperature-independent. Chamber walls stay at $T_0 = 300\,\mathrm{K}$. The measured net outward radiative power per area is $J = \varepsilon\sigma(T^4 - T_0^4)$, where $\sigma = 5.67 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$. Conduction through supports is negligible.

T/K	600	700	800	900	1000
$J/(\mathrm{kW\,m^{-2}})$	5.50	10.5	18.2	29.4	45.0

Each measured J has uncertainty $\pm 0.50\,\mathrm{kW\,m^{-2}}$. The graph uses $X = (T^4 - T_0^4)/(10^{11}\,\mathrm{K^4})$. A fit through the origin gives gradient $(4.54 \pm 0.08)\,\mathrm{kW\,m^{-2}}$ per unit X. Treat temperature uncertainties as negligible in parts(a)–(c).



(a) Calculate X for the $800\,\mathrm{K}$ measurement. Explain why a straight-line plot against X is appropriate. [2]

(b) Use the fitted gradient to determine the emissivity and its absolute uncertainty. [3]

(c) If the same data were plotted as J against T^4, determine the expected vertical intercept in $\mathrm{W\,m^{-2}}$ and explain its sign. [2]

1B Question 2 | Question paper

(d) For a tile at $800\,\mathrm{K}$, the chamber is cooled to $150\,\mathrm{K}$. Predict the fractional change in J. Explain why fitting $\log J$ against $\log T$ without a background correction need not yield exactly4. [3]

End of question paper

PHYSICS

Higher Level

Set A | Paper 1A and 1B

Worked solution book

Complete the examination before consulting this book. Marking points indicate a method (M), accuracy (A) or reasoning (R) award. The number after the code gives its value. Award only points supported by the candidate's retained working.

Accept valid equivalent approaches and reasonable rounding. Apply error carried forward only where the subsequent method remains physically meaningful; do not reward a contradiction of stated conditions. A correct multiple-choice letter earns one mark; no negative marks apply.

MCQs are automatically marked on the portal. Written marks are self-assessed. Reports show these separately and do not calculate an official IB grade.

Paper 1A - Answer key and explanations

1	2	3	4	5	6	7	8	9	10
B	C	B	A	A	C	C	A	A	A

11	12	13	14	15	16	17	18	19	20
B	D	D	C	D	B	C	D	D	D

21	22	23	24	25	26	27	28	29	30
C	D	B	B	B	C	A	B	A	C

31	32	33	34	35	36	37	38	39	40
B	D	A	B	D	C	C	D	A	A

1A Question 1 [1 mark]

Correct answer: B

The two roots of $4.4 = 12t - 5t^2$ differ by $\Delta t = \sqrt{12^2 - 2(10)(4.4)}/5 = 1.49666\,\text{s}$. Thus $\Delta x = 16\Delta t = 23.95\,\text{m}$. A halves this separation, D uses the time to the apex, and C uses the full return-to-launch time.

1A Question 2 [1 mark]

Correct answer: C

$a = (5^2 - 3^2)/(2 \times 4) = 2.0\,\text{m s}^{-2}$ and $v^2 = 9 + 2a(16) = 73$, giving $8.54\,\text{m s}^{-1}$. B incorrectly applies a square-root distance ratio to the speed increase rather than to squared speed; A applies a from-rest ratio to the full speed; D assumes speed itself grows linearly with position.

1A Question 3 [1 mark]

Correct answer: B

At terminal descent $b/m = g/v_t = 0.50\,\text{s}^{-1}$. Taking up as positive gives $\mathbf{a} = (-6, -10 + 8) = (-6, -2)\,\text{m s}^{-2}$, so its magnitude is $\sqrt{40} = 6.32$. A omits horizontal drag; C adds orthogonal contributions incorrectly; D confuses speed with acceleration.

1A Question 4 [1 mark]

Correct answer: A

Momentum gives $v_f = (2 \times 4 - 3 \times 1)/5 = 1.0\,\text{m s}^{-1}$ right. Initial energy is $17.5\,\text{J}$ and final energy $2.5\,\text{J}$, so the lost fraction is $15/17.5 = 6/7$. B is the retained fraction; C and D result from treating momentum or speeds as energy.

1A Question 5 [1 mark]

Correct answer: A

The downslope weight component is $mg\sin 30^\circ = 10\,\text{N}$. Friction is $\mu mg\cos 30^\circ = 5\,\text{N}$, so the required pull is $15\,\text{N}$. C ignores gravity, D ignores friction, and B uses the full weight as a downslope force.

1A Question 6 [1 mark]

Correct answer: C

The stored gravitational energy is $mgh = 16\,\text{kJ}$, so lowering returns $8.0\,\text{kJ}$. Net import is $25 - 8 = 17\,\text{kJ}$. D is recovered energy, B is upward loss only, and A adds recovered energy to the input.

1A Question 7 [1 mark]

Correct answer: C

$\alpha = (\tau - \tau_f)/I$. The slope is $12/6 = 2.0\,(\text{kg m}^2)^{-1}$, giving $I = 0.50\,\text{kg m}^2$. The horizontal intercept is $1.0\,\text{N m}$. D/B invert the slope incorrectly; A/B confuse the vertical intercept with a torque.

1A Question 8 [1 mark]

Correct answer: A

$\gamma = 5/3$ and lab flight time is $2.0\,\mu\text{s}$, so proper elapsed time is $1.2\,\mu\text{s}$. Survival is $e^{-0.6} = 0.549$. B multiplies laboratory time by the Lorentz factor, D uses laboratory time directly, and C divides by the Lorentz factor twice. The exponential must use elapsed time on the muons' own clocks.

1A Question 9 [1 mark]

Correct answer: A

$\gamma = 1.25$, so $\Delta x\prime = \gamma(900 - 0.60 \times 540) = 720\,\text{m}$. Also $c\Delta t\prime = 1.25(540 - 0.60 \times 900) = 0$: these events are simultaneous in $S\prime$. B/D omit a transformation term/factor; C transforms length without checking simultaneity.

1A Question 10 [1 mark]

Correct answer: A

Thermal resistances $L/(2kA)$ and $2L/(kA)$ are in ratio $1:4$. Therefore one fifth of the $100\,\mathrm{K}$ drop occurs across the left slab, leaving the interface at $80^\circ\mathrm{C}$. C reverses the order; D ignores different resistances; B uses thickness alone.

1A Question 11 [1 mark]

Correct answer: B

$(1-\alpha)/\varepsilon$ is initially 0.80 and finally $0.40/0.50 = 0.80$, so the equilibrium temperature is unchanged. D/A account for reduced absorption but omit the compensating emission change; C reverses that imbalance.

1A Question 12 [1 mark]

Correct answer: D

At top-of-atmosphere equilibrium the layer emits F upward and therefore F downward. The surface emits $2F$, so $T_s/T_0 = (2F/F)^{1/4} = 2^{1/4}$. B ignores downward radiation; C/A treat temperature as proportional to a square root/first power of flux.

1A Question 13 [1 mark]

Correct answer: D

A constant pressure offset does not change $dP/dT = nR/V$. Therefore
$V = 0.200(8.31)/400 = 0.004155\,\mathrm{m^3} = 4.16\,\mathrm{L}$. C/A misread the slope or amount; B omits division by the gradient.

1A Question 14 [1 mark]

Correct answer: C

Equal P, V, T imply equal particle numbers and $U = 3PV/2$, hence an internal-energy ratio of one. Since $mv_{\mathrm{rms}}^2 = 3k_BT$, the speed ratio is $1/\sqrt{2}$. A reverses the mass dependence; B/D confuse equal volumes with equal gas masses.

1A Question 15 [1 mark]

Correct answer: D

$\Delta U = \frac{3}{2}(P_CV_C - P_AV_A) = 1500\,\mathrm{J}$. Work by the gas is $P_A(V_B - V_A) = 400\,\mathrm{J}$. Thus
$Q = \Delta U + W = 1900\,\mathrm{J}$. A subtracts the work; B omits it; C overcounts the pressure-volume work.

1A Question 16 [1 mark]

Correct answer: B

$P_R = \mathcal{E}^2 R/(R+r)^2$. Its maximum occurs at $R = r$, since $(R+r)^2 \geq 4Rr$. At $R = 2r$, $P = 2\mathcal{E}^2/(9r)$, so $P/P_{\max} = 8/9$. C/A use the current or voltage ratio as a power ratio; D ignores internal losses and the matching condition.

1A Question 17 [1 mark]

Correct answer: C

The equilibrium is $x_e = 0.25\,\text{m}$. Writing $\xi = x - x_e$ gives $F = -24\xi$, so $\omega = \sqrt{24/0.600}$ and $T = 2\pi/\omega = 0.993\,\text{s}$. D confuses $1/\omega$ with T; A/B halve/double the correct period. The force offset changes equilibrium, not stiffness.

1A Question 18 [1 mark]

Correct answer: D

For a right-moving profile $y(x - vt)$, particle velocity has the opposite sign to the local slope. The slope at P is negative, so velocity is upward. Its displacement is positive, giving restoring acceleration downward. The other rows reverse one or both signs.

1A Question 19 [1 mark]

Correct answer: D

$\Delta\phi = \pi/2 - 2\pi(r_B - r_A)/\lambda = \pi/2 - \pi/2 = 0$. The sources arrive in phase. A ignores path delay, C adds delay with the wrong sign, and B ignores the specified source lead.

1A Question 20 [1 mark]

Correct answer: D

Geometrically possible orders are $n = 0, \pm1, \pm2, \pm3, \pm4$. An envelope minimum coincides when $m = nb/d = n/2$ is nonzero integer. Thus the even nonzero orders vanish, leaving $0, \pm1, \pm3$: five maxima. C omits the central maximum; A counts eight nonzero possible orders; B ignores the envelope.

1A Question 21 [1 mark]

Correct answer: C

The open-pipe spacing 90 Hz equals $v/(2L)$. C closed-open pipe has lowest frequency $v/(4L) = 45\,\text{Hz}$. A treats the two given frequencies as a closed-pipe sequence; B keeps the old fundamental; D takes the first listed resonance as fundamental.

1A Question 22 [1 mark]

Correct answer: D

More damping dissipates more energy per cycle, reducing the resonant amplitude and broadening the response. The frequency of peak displacement moves below the undamped natural frequency as damping grows. B/C reverse these trends; A omits the energy-loss effect. No quantitative damping formula is required.

1A Question 23 [1 mark]

Correct answer: B

$f_a/f_r = (v+u)/(v-u) = 2$, so $v+u = 2v-2u$ and $u = v/3$. A/D/C do not satisfy the two-sided frequency ratio. Only the source moves, so this remains within the current mechanical-Doppler scope.

1A Question 24 [1 mark]

Correct answer: B

The instantaneous power $\mathbf{F}\cdot\mathbf{v}$ is zero because force and velocity are perpendicular, so $dK/dt = 0$ at this instant. But $d\mathbf{p}/dt = \mathbf{F} \neq 0$. D/C ignore the change of direction; A wrongly assumes every force changes speed. This is an instantaneous result, not a claim of a permanently circular orbit.

1A Question 25 [1 mark]

Correct answer: B

Initially $E_i = -K$ and finally $E_f = -K/0.80 = -1.25K$. Thus dissipation is $E_i - E_f = 0.25K$, although the kinetic energy increases. D uses the fractional radius change as an energy change; A/C are unrelated retained/final-energy factors.

1A Question 26 [1 mark]

Correct answer: C

Between the charges their fields point in the same direction. To the left write the distance from $+Q$ as s; cancellation requires $Q/s^2 = 4Q/(s+3d)^2$, so $s+3d = 2s$ and $s = 3d$. Hence $x = -3d$. D gives unequal magnitudes, B lies between charges, and A is beyond the larger charge where its field dominates.

1A Question 27 [1 mark]

Correct answer: A

$V_S - V_R = +E(3.0) = 600\,\text{V}$, because potential increases against the field. The external work equals $q\Delta V = 1.2\,\mu\text{J}$. C/D use the full five-metre path; C/B also reverse the sign. Horizontal displacement adds no work.

1A Question 28 [1 mark]

Correct answer: B

$r = mv/(|q|B)$, so the SI gradient is $\Delta r/\Delta v = 0.40/(2.0 \times 10^5) = 2.0 \times 10^{-6}$ s. Thus $|q|/m = 1/(B\,\text{gradient}) = 1.25 \times 10^6\ \text{C kg}^{-1}$. A takes the reciprocal quantity; C/D mis-handle the graph scale or field.

1A Question 29 [1 mark]

Correct answer: A

Undeflected motion selects $v = E/B$, independent of mass and charge. In the second region $r = mv/(qB)$, so $r_\alpha/r_p = 4/2 = 2$. D reverses the ratio, C keeps only the common speed, and B neglects the alpha charge.

1A Question 30 [1 mark]

Correct answer: C

Outward flux increases as area increases. Lenz's law requires an inward induced field, hence clockwise current. Each current segment experiences an inward radial magnetic force, opposing expansion; the vector sum of these forces is zero. A reverses the stress; B/D reverse the induced field.

1A Question 31 [1 mark]

Correct answer: B

The signed integral is $-N(\Phi_f - \Phi_i) = +0.20\ \text{V s}$, giving $+0.0333\ \text{V}$. For the magnitude, total absolute flux change is $(6 + 10)\ \text{mWb}$, so the mean is $50(0.016)/6 = 0.133\ \text{V}$. C swaps the two averages, A misses Faraday's sign, and D does not time-average the total change.

1A Question 32 [1 mark]

Correct answer: D

At low scattering probability the number of target nuclei met is proportional to thickness, so the large-angle fraction approximately halves. Most atoms are largely empty space and most particles still pass nearly straight through. B reverses the target-number effect; C/A invent a uniform large energy loss.

1A Question 33 [1 mark]

Correct answer: A

Ionization from $n = 2$ needs $13.6/4\text{eV}$. The line energy is $13.6(1/4 - 1/9) = 13.6(5/36)\text{eV}$. Their ratio is $(1/4)/(5/36) = 9/5$. B reverses the ratio; D/C use an incorrect initial level or subtraction.

1A Question 34 [1 mark]

Correct answer: B

Photon energy doubles, so fixed optical power supplies half as many photons each second and the saturation current halves. If the original photon energy is E and work function $\Phi > 0$, $K_2 = 2E - \Phi = 2(E - \Phi) + \Phi > 2K_1$. D/C confuse power with photon rate; A reverses the work-function offset.

1A Question 35 [1 mark]

Correct answer: D

Compton shifts give final wavelengths $3\lambda_C$ at 180° and $2\lambda_C$ at 90°. Relative to initial photon energy E_i, the transferred energies are $2E_i/3$ and $E_i/2$. Their ratio is $4/3$. C reverses the comparison, A ignores angle, and B wrongly makes transferred energy proportional to the wavelength shift.

1A Question 36 [1 mark]

Correct answer: C

$K = qV$ and $p = \sqrt{2mK}$, so $p_Y/p_X = \sqrt{4 \times 2} = 2\sqrt{2}$. With $\lambda = h/p$, the wavelength ratio is $1/(2\sqrt{2})$. A/D omit the charge or misuse the square root; B reverses or incompletely applies the mass/charge dependence.

1A Question 37 [1 mark]

Correct answer: C

Writing raw rate as $B + S/2^n$, $B = 2(70) - 110 = 30\,\mathrm{s}^{-1}$ and $S = 80\,\mathrm{s}^{-1}$. At $3T$ the source contributes 10 and raw rate is 40. D is background alone, A halves a background-contaminated difference incorrectly, and B assumes a linear decrease.

1A Question 38 [1 mark]

Correct answer: D

At unchanged mass number, each beta-minus decay raises proton number by one and emits an electron plus antineutrino. Raising Z from 38 to 40 therefore requires two such decays. A/C/B have the wrong number or sign of charge change; the preceding fission conserves both A and Z.

1A Question 39 [1 mark]

Correct answer: A

From $L = 4\pi R^2 \sigma T^4$, $R_Y / R_X = \sqrt{100}(10000/5000)^2 = 40$. C reverses luminosity, B mishandles the temperature power, and D ignores temperature. A cooler yet more luminous star must have much larger area.

1A Question 40 [1 mark]

Correct answer: A

The measured sweep is $2p$, so $p = 0.040$arcsec and $d = 25$pc. Then $L = 4\pi d^2 b = 4.49 \times 10^{27}$ W. D uses the full sweep as parallax, C halves parallax a second time, and B fails to convert parsecs to metres.

Paper 1B - Worked solutions

1B Question 1 [10 marks]

(a) [2 marks]

M1 The lost apparent mass is $0.00480\,\text{kg}$ and buoyancy gives $(m_{air} - m_{immersed})g = \rho_l V g$.

A1 $\rho_l = 0.00480/(4.00 \times 10^{-6}) = 1200\,\text{kg}\,\text{m}^{-3}$.

(b) [2 marks]

R1 The plotted error bars are consistent with a straight line through the origin; this supports the proportional model over this measured range, without proving it universally.

A1 $1\,(\text{mm}\,\text{s}^{-1})/\text{mm}^2 = 10^3\,\text{m}^{-1}\,\text{s}^{-1}$, so $s = 1.20 \times 10^4\,\text{m}^{-1}\,\text{s}^{-1}$.

(c) [3 marks]

M1 $\eta = 2g(\rho_s - \rho_l)/(9s) = 1.199\,\text{Pa}\,\text{s}$.

M1 $\Delta\eta/\eta = 75/(7800 - 1200) + 0.4/12.0 = 0.044697$. The numerator is a density difference, so use its absolute uncertainty before dividing.

A1 $\eta = (1.20 \pm 0.05)\,\text{Pa}\,\text{s}$; an appropriately rounded equivalent such as 1.199 ± 0.054 is accepted.

(d) [3 marks]

M1 The largest bead has $r = \sqrt{3.24}\,\text{mm} = 1.8\,\text{mm}$ and $v_t = 0.0388\,\text{m}\,\text{s}^{-1}$, giving $\text{Re} = 0.140$.

R1 This is below one but not arbitrarily small; smaller beads reduce r and v_t, lowering Re and improving the creeping-flow approximation.

R1 Low Re alone does not prove terminal motion. Check that equal later time intervals give equal displacements, or that a late displacement-time trace has constant gradient; otherwise the force balance used for v_t is invalid.

1B Question 2 [10 marks]

(a) [2 marks]

A1 $X = (800^4 - 300^4)/10^{11} = 4.015$.

R1 The stated model has $J = (\varepsilon\sigma\, 10^{11})X$, giving a straight line through the origin if the model and constant emissivity apply.

(b) [3 marks]

M1 Convert the gradient to $4.54 \times 10^3\ \mathrm{W\,m^{-2}}$ and use $\varepsilon = s/(\sigma 10^{11})$.

A1 $\varepsilon = 0.800705$.

A1 $\Delta\varepsilon = \varepsilon(0.08/4.54) = 0.0141$, so $\varepsilon = 0.801 \pm 0.014$ or an appropriate equivalent.

(c) [2 marks]

A1 The extrapolated intercept is $-\varepsilon\sigma T_0^4 = -367.74\ \mathrm{W\,m^{-2}}$.

R1 It represents absorption of chamber radiation subtracted from the tile emission; the negative extrapolation at $T = 0$ is not negative gross emitted power.

(d) [3 marks]

M1 $J_{new}/J_{old} = (800^4 - 150^4)/(800^4 - 300^4) = 1.01891$.

A1 The net outward power increases by about 1.89%.

R1 J is proportional to $T^4 - T_0^4$, rather than T^4. Thus $\log J = \log(\varepsilon\sigma) + \log(T^4 - T_0^4)$ and the background term prevents an exact slope4 unless it is negligible or corrected.

End of worked solution book